Evaluating Normalized Bootstrap Gain-Variance Penalties for Robust Decision Tree Classification

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***Abstract*—This study evaluates whether theoretically motivated modifications of a stability-penalized decision tree (SPDT) improve classification performance, robustness to label noise, or split-selection stability. The original criterion scores a split as Gini gain minus a weighted bootstrap variance of that gain; we test a normalized variant that divides the penalty by $\mathrm{Gain}^2 + \varepsilon$, targeting the scale-dependence of the original penalty, and we benchmark against a faithful re-implementation of ANLS, a published node-level bootstrap stabilization method. All methods share identical repeated nested cross-validation splits over eight UCI datasets (45 outer evaluations per dataset). On clean data the normalized variant does not improve over custom CART (3 wins, 5 losses, mean Macro-F1 difference -0.0001) and is slightly below the original SPDT (2 wins, 6 losses); the Friedman omnibus test is not significant ($\chi^2(3) = 3.75$, $p = 0.290$). Under training label noise of 10–20%, an exploratory signal favors the normalized penalty (median paired differences up to $+0.0217$ with bootstrap confidence intervals excluding zero), which is not confirmed by the primary omnibus protocol. Split-selection stability does not improve (normalized variant has the lowest root-feature agreement, 0.78), and ANLS achieves the highest split stability yet not better prediction stability, empirically separating the two notions. Computational cost of the penalized trees is modest at refit time but tuning requires hundreds of bootstrap gain evaluations per split. We report positive, negative, and inconclusive findings without superiority claims.**

***Index Terms*—Decision trees, split criterion, bootstrap stability, gain variance, label noise, nested cross-validation.**

I. Introduction

DECISION trees are fast and interpretable, but their structure is unstable: small training perturbations change the selected split [1], motivating ensembles [2]–[4] and stabilization methods [5]–[9]. A previous experiment of ours evaluated a candidate criterion $\mathrm{Score}(s) = \mathrm{Gain}(s, D) - \lambda \mathrm{Var}_b[\mathrm{Gain}(s, D^{(b)})]$; descriptively it ranked best among single trees yet provided no confirmatory evidence of improvement (Friedman $p = 0.1353$), did not improve prediction stability, and cost roughly $40\times$ CART fitting time. That formulation's penalty is scale-dependent, which hinders the interpretation of λ.

This extension asks five questions. **RQ1:** does a normalized bootstrap gain-variance penalty improve Macro-F1 relative to CART and the original SPDT? **RQ2:** do SPDT or its normalized variant resist corrupted training labels better? **RQ3:** does penalizing bootstrap gain variance improve split-selection stability, even if prediction stability does not improve? **RQ4:** are any observed benefits worth the computational cost? **RQ5:** how does the method compare with the closest reproducible published stabilization method (ANLS [7])? Null hypothesis: no systematic difference across independent datasets.

II. Related Work

Tree stability measurement and stabilization have a long history: quantitative stability measures [5], tree similarity with node-level bootstrap stabilization (NLS) [6], [9], multi-attribute split tests [8], and ANLS, which iterates per-node bootstraps until a binomial Z-test confirms the modal predictor and uses the median winning cutoff [7]. Modified split criteria include Tsallis-entropy measures [10], label-noise-robust criteria [11], bias-corrected Gini selection [12], uncertainty-gated stream splitting [13], environment-consistent splitting [14], and minimum-split-gain regularization [15]. Coefficient-of-variation impurity measures [16] are keyword-adjacent but use variability as the *impurity*, never as a dispersion penalty on the gain. A prior-art search (2026-10-22) found no formula-level equivalent of $\mathrm{Gain} - \lambda \mathrm{Var}_b[\mathrm{Gain}]$ or its normalized form; a negative search is not proof of novelty, and both formulations remain candidates.

III. Proposed Methods

A. Original SPDT (baseline)

For node data D with Gini impurity

$$\mathrm{Gini}(D) = 1 - \sum_{c=1}^{C} p(c \mid D)^2, \tag{1}$$

the CART gain of a candidate split s is

$$\mathrm{Gain}(s, D) = \mathrm{Gini}(D) - \tfrac{|D_L|}{|D|}\,\mathrm{Gini}(D_L) - \tfrac{|D_R|}{|D|}\,\mathrm{Gini}(D_R). \tag{2}$$

With B bootstrap resamples $D^{(b)}$ of the node rows,

$$V(s) = \mathrm{Var}_b\Big[\mathrm{Gain}(s, D^{(b)})\Big], \qquad b = 1, \ldots, B, \tag{3}$$

$$\mathrm{Score}(s) = \mathrm{Gain}(s, D) - \lambda V(s). \tag{4}$$

Deterministic policies: midpoint thresholds between sorted distinct values; validity constraints on original counts; empty resample children contribute zero; seeded bootstrap per node. For λ=0 the estimator reduces to a custom CART reference verified against an independent implementation.

B. Normalized SPDT (candidate)

$$\mathrm{Score}_{\mathrm{norm}}(s) = \mathrm{Gain}(s, D) - \lambda \frac{V(s)}{\mathrm{Gain}(s, D)^2 + \varepsilon}, \qquad (5)$$

with $\varepsilon > 0$ stabilizing near-zero gains. The normalization makes the penalty scale-free with respect to the gain magnitude, addressing the main limitation of (4). This is an exploratory hypothesis; it is not claimed to be superior or novel.

C. ANLS Comparator

ANLS [7] repeatedly draws bootstrap resamples at each node, finds the plain CART-optimal split per replicate, and stops when the binomial Z-statistic of the top-two predictor frequencies exceeds Z_α (α=90%, minimum 10 replicates); the stabilized split uses the modal predictor with the median of its winning cutoffs. Our faithful re-implementation follows the paper's pseudocode with a documented iteration cap of 400 (the paper reports median 23 and mean 364 iterations per node); all datasets are numeric, so the categorical branch is omitted.

IV. Experimental Setup

Datasets: the same eight UCI tabular datasets as the baseline study (Table I); sources and SHA256 hashes are unchanged. Design: repeated stratified nested CV, outer 5 folds $\times$ 3 repeats on seeds $[42, 123, 2026]$ (45 evaluations per dataset–method, identical splits for all methods), inner stratified 3-fold tuning; median imputation fitted on training folds only.

Tuning budget (staged, documented). Stage S1 tunes the tree structure on plain custom CART ($4\times3\times3$ configurations); S2 and S3 tune $\lambda \in \{0.1, 1, 10\}$ for the original and normalized criteria at the frozen structure with B=10 and ε=10^{-4}. After a pilot on three datasets, the predefined budget was reduced and frozen: random forest context benchmark uses 100 trees; two noise realizations per level; eight stability subsamples; the ANLS iteration cap above. All decisions used training folds only.

Label noise protocol. Outer training labels are corrupted at levels $\{0, 5, 10, 20, 30\}\%$ (each corrupted label moves to a different class; deterministic seeds; identical corruption masks across methods; two realizations per level). Hyperparameters are frozen from clean-label inner CV; outer test folds remain clean. *Stability protocol.* Per outer fold, eight models are trained on 90% subsamples with frozen parameters; we measure pairwise root-feature agreement, root-threshold variation within the modal feature only, and prediction disagreement on the fixed outer test fold. The unit of inference is the dataset.

TABLE I
Dataset Inventory

Dataset	N	Features	Classes	Source
banknote	1372	4	2	ucimlrepo
glass	214	9	6	ucimlrepo
ionosphere	351	34	2	ucimlrepo
iris	150	4	3	ucimlrepo
seeds	210	7	3	UCI zip
sonar	208	60	2	ucimlrepo
wdbc	569	30	2	ucimlrepo
wine	178	13	3	ucimlrepo

V. Results

A. Baseline Recap (Separate Experiment)

The original study reports SPDT versus custom CART: 6 wins, 0 ties, 2 losses, mean Macro-F1 difference $+0.0058$; Friedman $\chi^2(2) = 4.0000$, $p = 0.1353$, Kendall's $W = 0.2500$ (not significant); SPDT prediction disagreement 0.1014 versus 0.0928 (custom CART); mean fitting time 21.43 s versus 0.53 s per outer fold. Those artifacts are preserved unchanged.

B. Clean-Data Comparison (Extension)

Table IX reports per-dataset Macro-F1 for six methods. The normalized variant does not improve over custom CART (3 wins, 5 losses, mean difference -0.0001) and is slightly below the original SPDT (2 wins, 6 losses, -0.0028). ANLS is descriptively the best single tree (5 wins, 3 losses, $+0.0066$), comparable in magnitude to the original SPDT's baseline gain. The Friedman omnibus over the four primary methods is not significant ($\chi^2(3) = 3.75$, $p = 0.290$), so confirmatory post-hoc tests were not run; paired analyses are exploratory with bootstrap confidence intervals.

VI. Robustness Analysis

Table X and Fig. 1 show Macro-F1 against training label noise. All methods degrade; the random forest remains the most robust reference. The normalized penalty degrades the least among single trees: median paired differences against custom CART are $+0.0063$ (5%), $+0.0136$ (10%, CI $[0.0015, 0.0186]$), $+0.0217$ (20%, CI $[0.0132, 0.0333]$), and $+0.0121$ (30%, CI $[-0.0051, 0.0280]$). Per-level Friedman tests are significant at 10% ($p = 0.024$) and 20% ($p = 0.003$) before family correction; after Holm correction across the five-level family only the 20% level remains significant ($p_{\mathrm{Holm}} = 0.017$). These noise analyses remain exploratory: they were not the primary confirmatory path, and the high-noise interval includes zero. We therefore report a promising but unconfirmed robustness signal concentrated at the 20% level, not a general robustness claim; a single pre-registered confirmatory level is left to future work.

VII. Stability Analysis

Table XI and Figs. 2–3 summarize stability. Penalizing bootstrap gain variance does *not* improve split selection: the normalized variant shows the lowest root-feature agreement

TABLE II
MACRO-F1 MEAN ± SD (DESCRIPTIVE FOLD-LEVEL SCORES) PER DATASET AND METHOD

Dataset	CART-skl	CART-cust	SPDT	RF	SPDT−CART
banknote	0.982 ± 0.008	0.981 ± 0.007	0.981 ± 0.007	0.994 ± 0.004	+0.0004
glass	0.620 ± 0.097	0.609 ± 0.093	0.626 ± 0.093	0.742 ± 0.075	+0.0166
ionosphere	0.873 ± 0.034	0.881 ± 0.038	0.879 ± 0.038	0.929 ± 0.037	-0.0027
iris	0.949 ± 0.033	0.945 ± 0.038	0.944 ± 0.037	0.955 ± 0.034	-0.0008
seeds	0.897 ± 0.032	0.906 ± 0.036	0.913 ± 0.034	0.923 ± 0.038	+0.0066
sonar	0.719 ± 0.065	0.706 ± 0.073	0.725 ± 0.061	0.833 ± 0.065	+0.0189
wdbc	0.925 ± 0.031	0.925 ± 0.030	0.928 ± 0.030	0.956 ± 0.023	+0.0037
wine	0.905 ± 0.047	0.905 ± 0.044	0.909 ± 0.046	0.981 ± 0.024	+0.0040

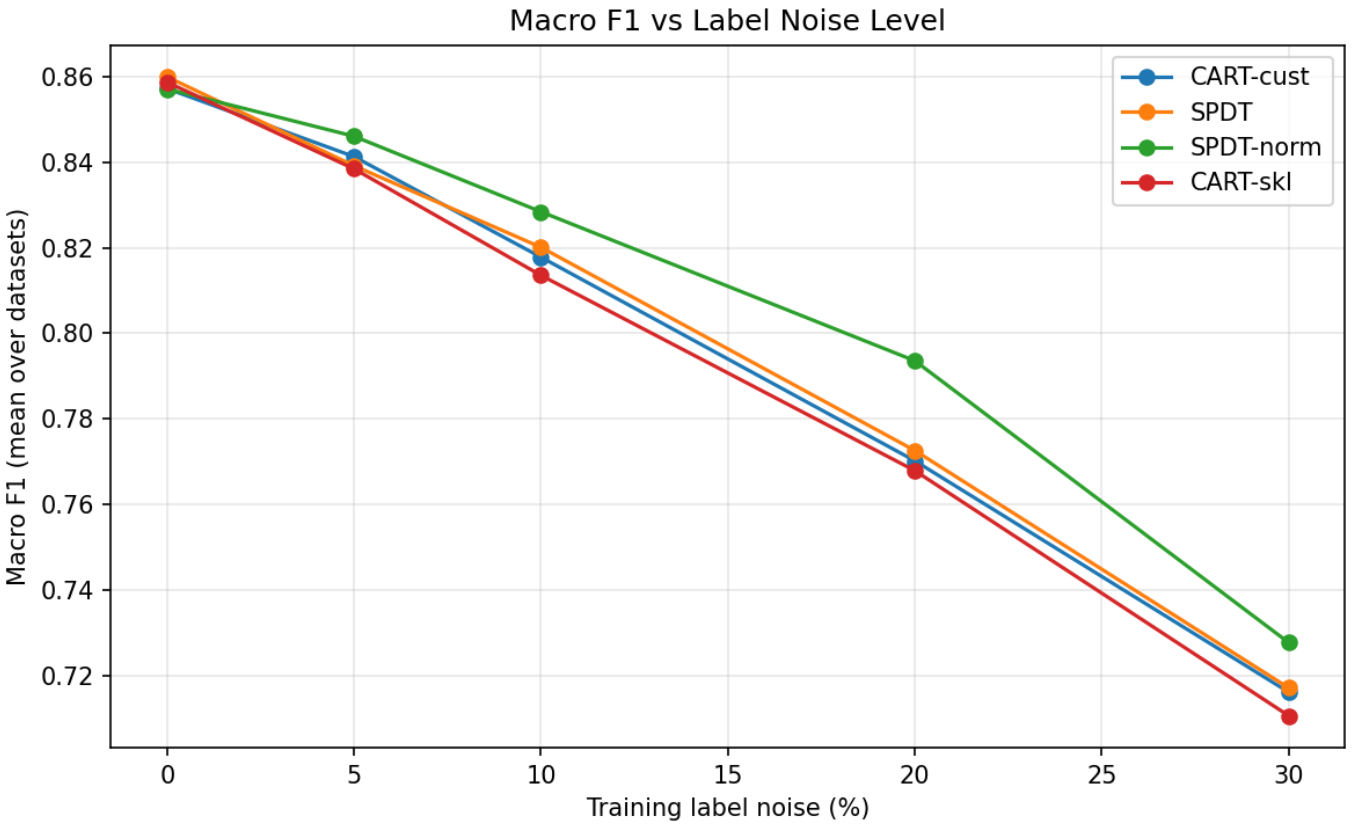


Fig. 1. Mean Macro-F1 versus training label noise level (four primary methods plus contextual random forest). Exploratory analysis.

TABLE III
MEAN MACRO-F1 RANK ACROSS DATASETS

Method	Mean rank
SPDT	1.500
CART-skl	2.000
CART-cust	2.500

TABLE IV
DATASET-LEVEL STATISTICAL TEST (FRIEDMAN, D=8 DATASETS, 3 METHODS)

Statistic	Value
χ^2	4.0000
Degrees of freedom	2
p	0.1353
Kendall's W	0.2500
Decision	post-hoc not run

(0.78 versus 0.83 for CART variants) and its prediction disagreement (0.105) exceeds both CART (0.093) and the original SPDT (0.096). ANLS attains the highest split agreement (0.85, consistent with its design) yet its prediction disagreement is the highest among single trees (0.109): stable splits do not guarantee stable predictions, and the two notions must be reported separately. Root-threshold variation is small for all methods and is only computed within the modal feature.

TABLE V
COMPUTATIONAL COST (PER OUTER FOLD, AVERAGED OVER DATASETS)

Method	Fit (s)	Fit median (s)	Predict (s)	Nodes	Depth
CART-cust	0.5340	0.4163	0.0009	26.6722	5.1000
CART-skl	0.2693	0.2605	0.0010	27.4222	5.2917
RF	1.1169	1.0863	0.0573	-	-
SPDT	21.4302	14.6491	0.0010	28.5944	5.4611

TABLE VI
PREDICTION STABILITY (MEAN PAIRWISE DISAGREEMENT; NOT AN ACCURACY METRIC)

Method	Mean disagreement	SD
CART-cust	0.0928	0.0853
CART-skl	0.0983	0.0885
RF	0.0271	0.0337
SPDT	0.1014	0.0934

TABLE VII
ABLATION ON IDENTICAL OUTER SPLITS

Variant	Macro F1	SD	Fit (s)	Nodes
A0 (λ=0)	0.8599	0.1270	0.0201	37.6611
A1 (λ=.1)	0.8598	0.1256	0.0860	37.6111
A1 (λ=1)	0.8601	0.1255	0.0862	37.5278
A2 (B=20)	0.8601	0.1251	0.1136	37.4556
A2 (B=5)	0.8613	0.1243	0.0711	37.4667
A3 d3 l1	0.8357	0.1592	0.0371	12.8722
A3 d3 l5	0.8353	0.1546	0.0373	11.8611
A3 d8 l1	0.8592	0.1272	0.0821	35.2500
A3 d8 l5	0.8521	0.1381	0.0820	24.7389

TABLE VIII
SPDT HYPERPARAMETER SELECTION FREQUENCY (45 OUTER FOLDS PER DATASET)

Parameter	Value	Frequency	Share
max_depth	5.0	140	0.3889
max_depth	8.0	102	0.2833
max_depth	3.0	92	0.2556
min_samples_leaf	1.0	201	0.5583
min_samples_leaf	2.0	97	0.2694
min_samples_leaf	5.0	62	0.1722
min_samples_split	2.0	252	0.7000
min_samples_split	5.0	69	0.1917
min_samples_split	10.0	39	0.1083
n_bootstrap	5.0	191	0.5306
n_bootstrap	20.0	87	0.2417
n_bootstrap	10.0	82	0.2278
penalty_lambda	10.0	233	0.6472
penalty_lambda	0.0	61	0.1694
penalty_lambda	0.01	41	0.1139
penalty_lambda	1.0	22	0.0611
penalty_lambda	0.1	3	0.0083

VIII. ABLATION STUDY

Table XII (identical splits, frozen structure) shows that all penalty configurations cluster within ± 0.013 Macro-F1; the normalized variant with small λ=0.1 ranks highest (0.8617) while large λ or small ε hurts slightly; depth restrictions (depth 3, 0.835–0.836) matter far more than any penalty setting.

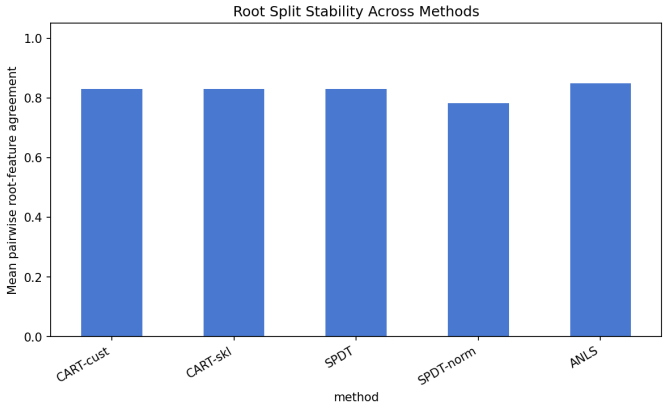


Fig. 2. Mean pairwise root-feature agreement over folds and datasets.

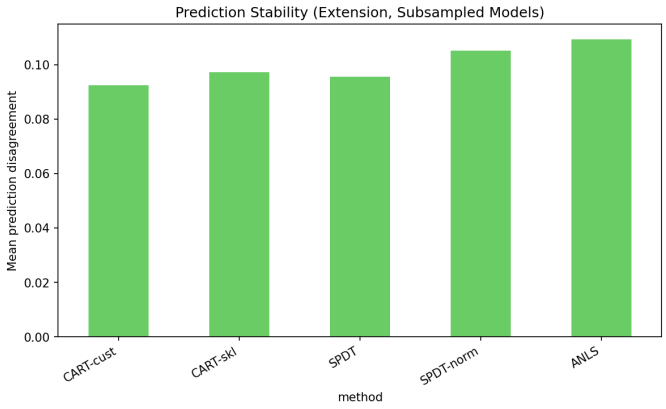


Fig. 3. Mean prediction disagreement of subsampled models.

Ordinary tree-complexity control, not the variance penalty, dominates the outcome variance.

IX. COMPUTATIONAL COST

Table XIII reports refit times of tuned configurations and counters from the estimators. A penalized refit evaluates about 18k candidate splits and, when $\lambda > 0$, roughly 178k bootstrap gain evaluations per outer fold; refit times remain under 0.1 s, ANLS averages 2.2 s, and a single-process memory probe measured 2.3 MB peak for a penalized fit. These figures exclude the staged tuning overhead (S1–S3, tens of inner-CV fits per outer fold), which is the dominant cost of the penalized methods and must be budgeted in practice. Extension timings report refits of tuned configurations, whereas the baseline study reported full inner-grid search times; the two bases differ and cross-version timing comparisons are not controlled. Gains of the size observed do not justify this overhead for clean-data deployment; the judgment is closer under label noise, where the exploratory signal appeared.

X. DISCUSSION

RQ1: the normalized penalty does not improve clean-data Macro-F1; it performs at or slightly below CART and the original SPDT, and the omnibus test finds no differences ($p =$

TABLE IX
MACRO-F1 (MEAN OVER 45 OUTER FOLDS) PER DATASET, EXTENSION EXPERIMENT A

Dataset	CART-cust	SPDT	SPDT-norm	ANLS	CART-skl	RF
banknote	0.9807	0.9810	0.9856	0.9774	0.9816	0.9932
glass	0.6095	0.6186	0.6160	0.6498	0.6200	0.7572
ionosphere	0.8813	0.8762	0.8776	0.8793	0.8728	0.9274
iris	0.9449	0.9456	0.9433	0.9412	0.9493	0.9546
seeds	0.9063	0.9078	0.9025	0.9116	0.8971	0.9218
sonar	0.7057	0.7216	0.7096	0.7118	0.7188	0.8221
wdbc	0.9247	0.9239	0.9213	0.9248	0.9247	0.9572
wine	0.9047	0.9051	0.9010	0.9143	0.9053	0.9804

TABLE X
MEAN MACRO-F1 BY TRAINING LABEL NOISE LEVEL

Noise	CART-cust	SPDT	SPDT-norm	RF
0%	0.8572	0.8600	0.8571	0.9142
5%	0.8413	0.8392	0.8461	0.9043
10%	0.8178	0.8201	0.8284	0.8948
20%	0.7702	0.7726	0.7936	0.8597
30%	0.7161	0.7171	0.7277	0.7914

TABLE XI
SPLIT AND PREDICTION STABILITY (MEAN OVER FOLDS AND DATASETS)

Method	Root-feature agr.	Threshold var.	Prediction dis.
CART-cust	0.8306	2.5564	0.0926
CART-skl	0.8301	2.5511	0.0972
SPDT	0.8296	2.6105	0.0957
SPDT-norm	0.7822	3.3918	0.1052
ANLS	0.8486	2.4415	0.1094

TABLE XII
ABLATION ON IDENTICAL SPLITS (FROZEN STRUCTURE, MEAN MACRO-F1)

Variant	Macro F1	SD
A3n_lam0.1	0.8617	0.1257
A2_B5	0.8586	0.1278
A4_d8_leaf1	0.8585	0.1283
A2_B20	0.8584	0.1293
A1_lam10	0.8583	0.1297
A1_lam1	0.8578	0.1308
A1_lam0.1	0.8577	0.1301
A0_lam0	0.8572	0.1314
A3n_lam10	0.8537	0.1323
A2n_eps0.001	0.8527	0.1297
A4_d8_leaf5	0.8521	0.1381
A2n_eps1e-06	0.8494	0.1238
A4_d3_leaf1	0.8359	0.1591
A4_d3_leaf5	0.8353	0.1546

TABLE XIII
COMPUTATIONAL COST PER OUTER FOLD (REFIT OF TUNED CONFIGURATION)

Method	Fit (s)	Predict (s)	Candidates	Boot. gain evals
ANLS	2.2063	0.0010	-	-
CART-cust	0.0173	0.0008	17727	0
CART-skl	0.0040	0.0009	-	0
RF	0.1550	0.0342	-	-
SPDT	0.0676	0.0009	17753	177531
SPDT-norm	0.0758	0.0009	19291	192914

0.290). A depth-matched control (custom CART restricted to the tuned depth, identical splits) shows that this is not purely a depth effect: the control averages 0.0021 *below* the original SPDT and is statistically indistinguishable from the normalized variant (+0.0007), so the original penalty carries a small effect beyond complexity restriction while the normalized one does not. **RQ2:** under 10–20% training label noise the normalized variant degrades least among single trees, with paired confidence intervals excluding zero; this exploratory signal requires independent confirmation before any robustness claim. **RQ3:** neither the original nor the normalized penalty improves split-selection stability; the normalized variant is the least stable at the root split, and prediction stability mirrors the baseline finding (slightly worse than CART). ANLS confirms that split stability and prediction stability are distinct quantities. **RQ4:** refit-time cost is modest, but tuning overhead is substantial; the observed clean-data effects do not justify the cost, while the noise-robustness signal is the only result that begins to do so. **RQ5:** ANLS, faithfully re-implemented, is descriptively the strongest single-tree comparator and a more promising stabilization avenue than gain-variance penalties on these datasets.

XI. LIMITATIONS

Eight small datasets limit statistical power; Gini impurity only; numeric features only; a single disagreement-based stability proxy; noise injected independently of features (real label noise is often not uniform); frozen clean-label tuning under noise is one of several defensible protocols; ANLS uses a capped iteration budget and no official code release exists to cross-validate our re-implementation; the closest alternative criteria (minimax split, conditional inference trees) were analyzed but not benchmarked; and the novelty of both penalty formulations remains unverified at the formula level.

XII. CONCLUSION

Across eight UCI datasets with leakage-free repeated nested CV, a normalized bootstrap gain-variance penalty did not improve clean-data classification over CART or the original SPDT, did not improve split or prediction stability, and adds tuning overhead. Its one promising signal—slower degradation

under 10–20% training label noise—is exploratory and awaits confirmatory replication on larger benchmarks. A faithful ANLS implementation was descriptively the best single-tree stabilized method, and empirically separated split stability from prediction stability. We conclude that, under the tested conditions, simple CART remains preferable for clean data, and that future stabilization research should prioritize voting-based per-node stabilization over gain-variance penalties.

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